

Transition to the over-the-barrier continuum in a superconducting two-photon detector

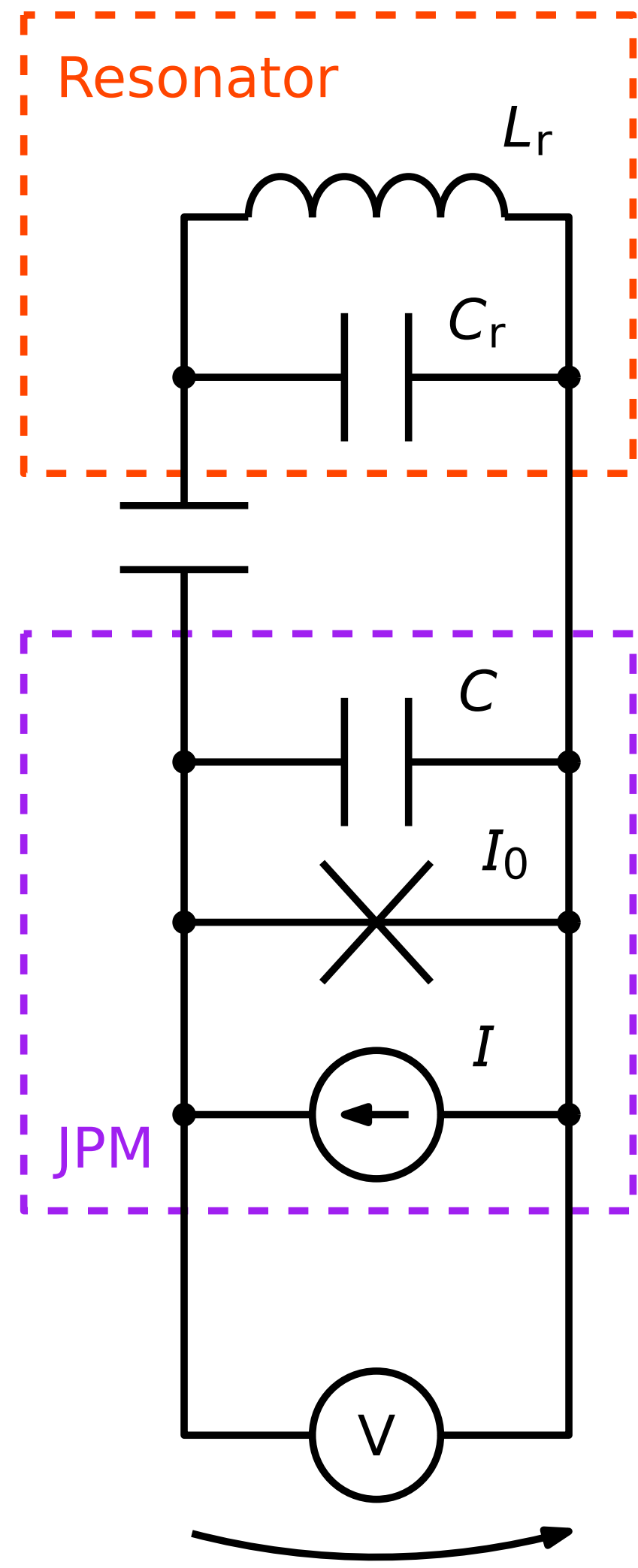
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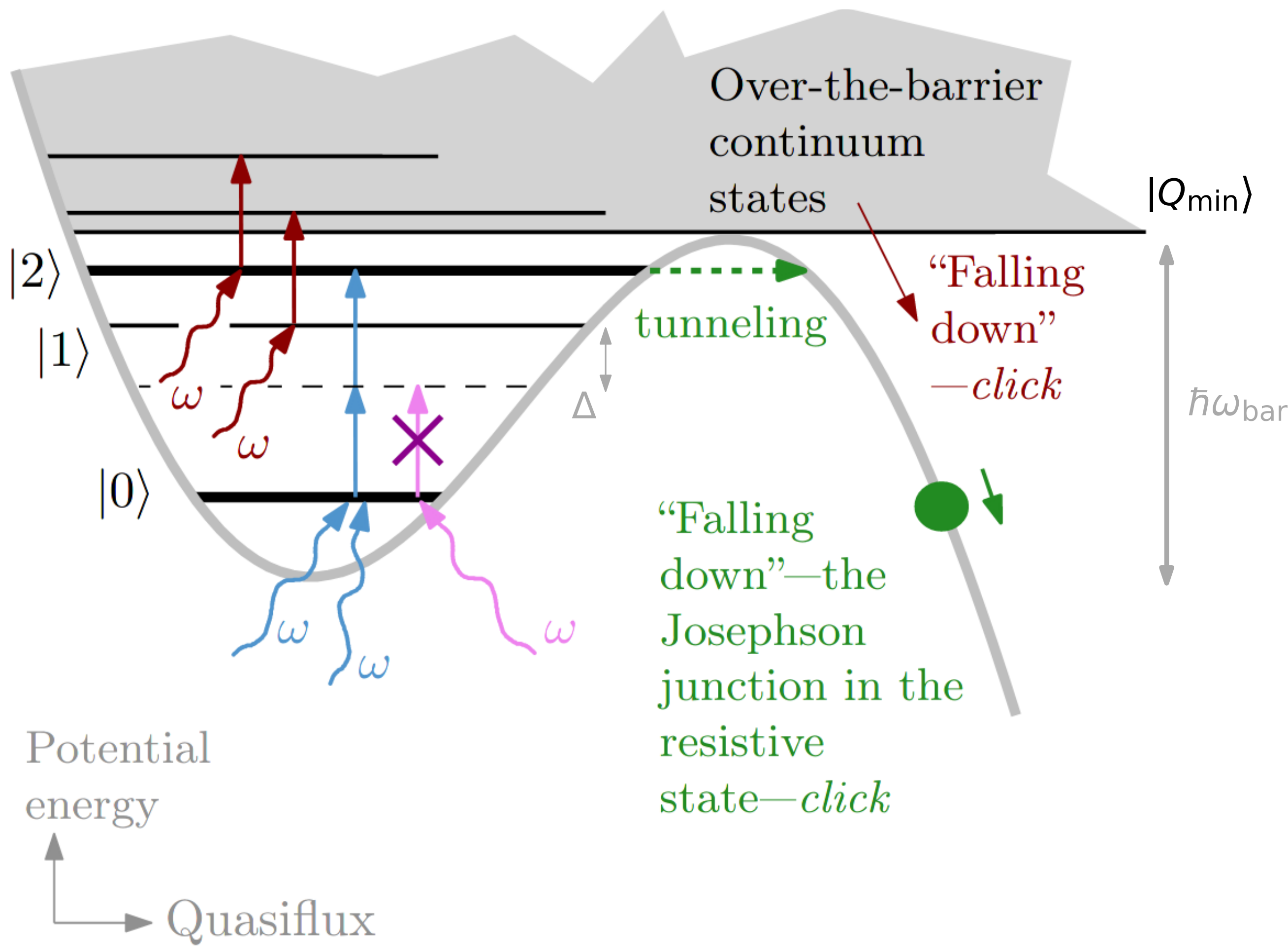
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The system, its model, and some dressing

Circuit



How Josephson photomultiplier (JPM) excites



More about the two-photon JPM: Phys. Rev. Appl. 14(6) 064063

How to describe the red excitations?

$$H = H_{\text{det}} + H_r + H_{\text{coup}}$$

$$H_{\text{det}} = \hbar(\omega + \Delta) |1\rangle \langle 1| + \hbar 2\omega |2\rangle \langle 2|, \quad H_r = \hbar\omega_r a^\dagger a$$

$$H_{\text{coup}} = H_1 + H_2 + H_{01} + H_{02}$$

$$H_1 = \int_{Q_{\min}}^{\infty} (R_{1Q} |Q\rangle \langle 1| a + \text{H. c.})$$

$$H_2 = \int_{Q_{\min}}^{\infty} (R_{2Q} |Q\rangle \langle 2| a + \text{H. c.})$$

H_{01}, H_{02} — describe $0 \rightarrow 1$ and $1 \rightarrow 2$ that we don't care about here

$$R_{nQ} = \frac{\tilde{C}'}{CC_r} \sqrt{\frac{\hbar}{2\rho}} \langle Q | Q | n \rangle, \text{ where } \rho = \sqrt{\frac{L_r}{C_r}}$$

Eliminating |1>

Dressed picture: $0 \rightarrow 2$ two-photon transition, $|1\rangle$ is excluded from H

Shifted JPM frequencies:

$$\tilde{\omega}_1 = \omega + \Delta + \chi_1, \quad \tilde{\omega}_2 = 2\omega - \chi_2$$

$$\chi_1 = g_1^2/\Delta, \quad \chi_2 \approx 2\chi_1, \quad g_1 = \frac{\tilde{C}'}{CC_r} \sqrt{\frac{\hbar}{2\rho}} \langle 1 | Q | 0 \rangle$$

H_1, H_2 obtain additional transitions to continuum

Derivation of the master equation

Fock state of the resonator $|\psi\rangle = \sum_{N=0}^{\infty} (\alpha_{N0} |N0\rangle + \alpha_{N1} |N1\rangle + \alpha_{N2} |N2\rangle) + \int_{Q_{\min}}^{\infty} dQ \beta_{NQ} |NQ\rangle$

Metastable state of the JPM

plug in to the Hamiltonian and obtain the Langevin equations for α_{Nm} by integrating out β_{NQ}

Approximations

- Markov approximation: $\kappa_{n1}, \kappa_{n2} \approx \text{const}$ around the Q s matching $\omega'_{N+11}, \omega''_{N+21}$, and ω'_{N+12}
- α -s change slowly
- We excite the states far enough from the barrier top

$$\kappa_{nm}(Q) = \frac{\pi}{\hbar^2} \sqrt{\frac{\hbar C}{2}} \frac{R_{nQ}^* R_{mQ}}{\sqrt{\omega_Q}}, \quad \omega_Q = \frac{Q^2}{2\hbar C}$$

Excitation height

$$Q'_{Nn} = \sqrt{2\hbar\omega'_{Nn}C}, \quad Q''_{Nn} = \sqrt{2\hbar\omega''_{Nn}C}$$

$$\omega'_{N1} = \tilde{\omega}_1 + \omega - \chi_1 N$$

$$\omega''_{N1} = \tilde{\omega}_1 + 2\omega - \chi_1 N$$

$$\omega'_{N2} = \tilde{\omega}_2 + \omega + 2\chi_1 N$$

Density matrix

$$\rho_{NnMm} = \alpha_{Nn}^* \alpha_{Mm}$$

The master equation

Fast tunneling $\longrightarrow$ Fast JPM decoherence
only the red and the brown terms survive

$$\dot{\rho}_{N0,N0} = \lambda_1 \frac{\gamma_{N-11}}{2} (\rho_{N-11,N0} e^{i(\Delta+2\chi_1)t} + \text{c. c.})$$

$$\dot{\rho}_{N-10,N0} = -\frac{\gamma_{N-11}}{2} \rho_{N-11,N0} \Theta(\omega'_{N-11} - \omega_{\text{bar}}) + \lambda_2 \frac{\gamma_{N-22}}{2} \sqrt{N-1} \rho_{N-22,N0} e^{-i(\Delta-(3N-8)\chi_1)t} + \lambda_1 \frac{\gamma_{N-11}}{2} \sqrt{N} \rho_{N-11,N-11} e^{-i(\Delta+2\chi_1)t}$$

$$\dot{\rho}_{N-11,N-11} = -\gamma_{N-11} \rho_{N-11,N-11} \Theta(\omega'_{N-11} - \omega_{\text{bar}}) + \lambda_2 \frac{\gamma_{N-22}}{2} \sqrt{N-1} \rho_{N-22,N-11} (e^{-i(\Delta-(3N-8)\chi_1)t} + \text{c. c.})$$

$$\dot{\rho}_{N-22,N-11} = -\frac{\gamma_{N-22} + \gamma_{N-11}}{2} \rho_{N-22,N-11} - \lambda_2 e^{i\Delta t} [\eta_{N-11}^* e^{-i(3N-8)\chi_1 t} + \frac{\gamma_{N-11}}{2} \sqrt{N-1} e^{-i(N-4)\chi_1 t} \Theta(\omega'_{N-11} - \omega_{\text{bar}})] \rho_{N-11,N-11} + \lambda_2 \frac{\gamma_{N-22}}{2} \sqrt{N-1} \rho_{N-22,N-22} e^{i(\Delta-(3N-8)\chi_1)t}$$

$$\dot{\rho}_{N-22,N-22} = -\gamma_{N-22} \rho_{N-22,N-22} - \lambda_2 \{ e^{i\Delta t} [\eta_{N-11}^* e^{-i(3N-8)\chi_1 t} + \frac{\gamma_{N-11}}{2} \sqrt{N-1} e^{-i(N-4)\chi_1 t} \Theta(\omega'_{N-11} - \omega_{\text{bar}})] \rho_{N-11,N-22} + \text{c. c.} \}$$

Interpretation

Single-photon excitation $|n\rangle \rightarrow |Q'_{Nn}\rangle$:

rates: $\gamma_{N1} = \kappa_{11}(Q'_{N1})N$ $\gamma_{N2} = \kappa_{22}(Q'_{N2})N$
escape (if possible) from |1> escape from |2>

Two-photon virtual excitation $|n\rangle \rightarrow |Q''_{Nn}\rangle$:

flow of coherence from |2> to |1> with the modulated rate $\eta_{N1} = \kappa_{21}(Q''_{N1})(N-1)\sqrt{N}$